```
-- Step 1 Create database
    show databases;
    use retail_store_db;

-- Step 2 Create table --
   create table sales;

-- Step 3: Insert Data --
insert into sales values
    (1,'Akash','Male',24,'Pune','Electronics','Laptop','UPI',1,55000,4.8,'2026-01-10'),
    (2,'Rahul','Male',26,'Mumbai','Fashion','Shirt','Card',2,2500,4.2,'2026-01-11'),
    (3,'Sneha','Female',23,'Delhi','Beauty','Lipstick','UPI',3,1500,4.5,'2026-01-12'),
    (4,'Priya','Female',25,'Pune','Electronics','Mobile','Card',1,25000,4.7,'2026-01-13'),
    (5,'Aman','Male',28,'Indore','Sports','Cricket Bat','Cash',1,3000,4.1,'2026-01-14'),
    (6,'Rohit','Male',27,'Mumbai','Electronics','Headphones','UPI',2,4000,4.3,'2026-01-15'),
    (7,'Riya','Female',24,'Delhi','Fashion','Jeans','Card',1,2200,4.4,'2026-01-16'),
    (8,'Karan','Male',29,'Pune','Sports','Football','Cash',2,1800,4.0,'2026-01-17'),
    (9,'Neha','Female',22,'Bhopal','Beauty','Face Wash','UPI',4,800,4.6,'2026-01-18'),
    (10,'Vikas','Male',31,'Jaipur','Electronics','Tablet','Card',1,18000,4.5,'2026-01-19'),
    (11,'Pooja','Female',27,'Pune','Fashion','Kurti','UPI',2,1700,4.3,'2026-01-20'),
    (12,'Arjun','Male',30,'Delhi','Electronics','Smart Watch','Card',1,9000,4.6,'2026-01-21'),
    (13,'Anjali','Female',26,'Mumbai','Beauty','Perfume','Cash',1,3500,4.2,'2026-01-22'),
    (14,'Yash','Male',25,'Indore','Sports','Badminton Kit','UPI',1,2800,4.4,'2026-01-23'),
    (15,'Meera','Female',28,'Pune','Electronics','Camera','Card',1,45000,4.9,'2026-01-24'),
    (16,'Sahil','Male',24,'Bhopal','Fashion','Jacket','UPI',1,3200,4.1,'2026-01-25'),
    (17,'Nikita','Female',29,'Jaipur','Beauty','Moisturizer','Card',2,1200,4.7,'2026-01-26'),
```

```
(18,'Harsh','Male',32,'Delhi','Sports','Gym Gloves','Cash',3,900,4.0,'2026-01-27'),
(19,'Simran','Female',23,'Mumbai','Electronics','Earbuds','UPI',2,2500,4.5,'2026-01-28'),
(20,'Deepak','Male',27,'Pune','Fashion','T-Shirt','Card',3,1800,4.2,'2026-01-29');

-- Queries
-- Q1. Display all records from the sales table --
select * from sales;

-- Q2. Display only customer names.--
select customer_name from sales;

-- Q3. Display customer names and cities. --
select customer_name, city from sales;

-- Q4. Display product names and prices.--
select product_name,price from sales;

-- Q5. Display all unique cities --
select distinct city from sales;

-- Q6. Display all unique categories --
select distinct category from sales;

-- Q7. Display all unique payment methods --
select distinct payment_mode from sales;
```

```
-- Q8. Display all male customers. --
    select * from sales where gender = 'male';

-- Q9. Display all female customers. --
    select * from sales where gender = 'female';

-- Q10. Display customer name and rating --
   select customer_name,rating from sales;

---- Part B - Where Clause--
-- Q11. Display all customers from Pune --
   select * from sales where city = 'pune';

-- Q12. Display all customers from Mumbai. --
          select * from customer where city = 'mumbai';

-- Q13 Display all Electronics products.
   select * from sales where category = 'electronics';

-- Q14. Display all Fashion products --
  select * from sales where category = 'fashion';

-- Q15. Display all Sports products.
   select * from sales where category = 'sports';
```

```
-- Q16. Display products with price greater than 5000.
   select product_name from sales where price >=5000;

-- Q17. Display products with price less than 5000.
   select product_name,price from sales where price<=5000;

--  Q18. Display customers whose age is greater than 25.
   select customer_name from sales where age >=25;

-- Q19. Display customers whose age is less than 25
   select customer_name from sales where age <=25;

-- Q20. Display products with rating greater than 4.5
          select product_name from sales where rating > 4.5;

-- Q21. Display all UPI transactions.
 select * from sales where Payment_mode = 'UPI';

-- Q22. Display all Card transactions.
 select * from sales where payment_mode = 'card';

-- Q23. Display all Cash transactions.
 select * from sales where payment_mode = 'cash';

-- Q24. Display Electronics products from Pune.
 select * from sales where category = 'electronics' or city = 'pune';
```

```
-- Q 25. Display Fashion products from Mumbai.--
 select * from sales where category = 'fashion' and city ='mumbai';

-- Q 26. Display Female customers from Delhi.--
  select * from sales where gender = 'female' and city = 'delhi';

-- Q27. Display Male customers from Pune --
  select * from sales where gender = 'male' and city = 'pune';

-- Q28. Display products priced between 2000 and 10000.--
  select * from sales where price between 2000 and 10000;

-- Q29. Display customers aged between 24 and 30. --
  select * from sales where age between 24 and 30;

-- Q30. Display orders placed after 20 January 2026 --
   select * from sales where order_date > '2016-01-20';

        -- Part C – ORDER BY, LIMIT and OFFSET --
-- Q31. Sort products by price in ascending order.--
  select * from sales order by price asc;

-- Q32. Sort products by price in descending order--
   select * from sales order by price desc;
```

```
-- Q 33. Sort customers by age in ascending order. --
  select * from sales order by age asc;

-- Q34. Sort customers by age in descending order --
   select * from sales order by age desc;

-- Q35. Sort products by rating in descending order.--
    select * from sales order by rating desc;

-- Q36. Display the highest priced product --
  select product_name,price from sales where price = (select max(price) from sales);

-- Q37. Display the lowest priced product.--
  select * from sales where price = (select(min(price) from sales);

-- Q38. Display the highest rated product --
  select max(rating) from sales;

-- Q39. Display the latest order --
  select * from sales order by Order_date desc limit 1;

-- Q40. Display the oldest order.--
  select * from sales order by order_date asc limit 1;

-- Q41. Display the top 5 records --
  select * from sales order by order_date asc limit 5 ;
```

```
-- Q42. Display the top 3 most expensive products --
  select product_name,price from sales order by price desc limit 3;

-- Q43. Display the top 5 highest rated products --
  select product_name, rating from sales order by price desc limit 5;

-- Q44. Display the second highest priced product. --
  select product_name,price from sales order by price desc limit 1 offset 1;

-- Q45. Display the third highest priced product --
  select product_name,price from sales order by price desc limit 1 offset 3;

-- Q46. Skip the first 5 rows and display the next records.--
  select * from sales limit 100 offset 5;

-- Q47. Display rows 6 to 10.--
  select * from sales limit 5 offset 5;

 -- Q48. Display rows 11 to 15. --
   select * from sales limit 10 offset 5;

-- Q49. Display rows 16 to 20 --
  select * from sales limit 5 offset 15;
```

-- Q50. Display the top 3 latest orders.--

```
select * from sales order by order_date desc limit 3
```

-- Part D – Aggregate Functions --

-- Q51. Find the total number of orders --

```
select count(order_date) from sales;
```

-- Q52. Find the total sales revenue--

```
select sum(price) as totalrevenue from sales;
```

-- Q53. Find the average sales amount. --

```
select avg(price) from sales;
```

-- Q54. Find the maximum sales amount.--

```
select max(price) from sales;
```

-- Q55. Find the minimum sales amount.--

```
select min(price) from sales;
```

-- Q56. Find the highest rating.--

```
select max(rating) from sales;
```

-- Q57. Find the loweest rating.--

```
select min(rating) from sales;
```

```
-- Q58. Find the average rating. --
    select avg(rating) from sales;

-- Q59 select count(electronis) from sales;
   select count(electronis) from sales

-- Q60. Find the average rating.--
  select avg(rating) from sales;

-- Q59. Find the total quantity sold.
  select count(category) from sales;

-- Q60. Find the average quantity sold.
  select avg(category) from sales;

-- Q61. Find the maximum quantity sold.
  select max(category) from sales;

-- Q62. Find the minimum quantity sold.
   select min(category) from sales;

-- Q63. Count the number of Electronics products --
  select count(Electronics) as Elec_count from sales where category = 'Electronics';

-- Q64. Count the number of Fashion products.--
  select count(fashion) as fashion_count from sales where category = 'fashion';
```

-- Q65. Count the number of Sports products --

```
select count(sports) as sport_count from sales where category = 'sports';
```

-- Part E – String Functions --

-- Q66. Convert all customer names to uppercase.--

```
select customer_name,upper(customer_name) as updated_names from sales;
```

-- Q67. Convert all customer names to lowercase --

```
select customer_name,lower(customer_name) as updated_names2 from sales;
```

-- Q Q68. Find the length of each customer name --

```
SELECT customer_name, LENGTH(customer_name) AS length_of_names FROM sales;
```

-- Q69. Find the length of each city name --

```
select city, length(city) as length_of_city from sales;
```

-- Q70. Combine customer name and city in one column. --

```
select concat(customer_name,'-',city) as customer_city from sales;
```

-- Q71. Replace the letter 'a' with '@' in customer names.--

```
select customer_name,replace(customer_name,'a','@') from sales;
```

-- Q72. Replace the letter 'e' with '*' in customer names.--

```
select customer_name,replace(customer_name,'e','*') from sales;
```

-- Q73. Remove extra spaces using TRIM() --

```
select customer_name,trim(customer_name) as cleaned_name from sales;
```

-- Q74. Display the first 3 letters of each customer name--

```
select customer_name, left(customer_name,3) as first_three_ltrs from sales;
```

-- Q75.Display the last 2 letters of each city name --

```
select city, right(city,2) as cities_last_2_ltrs from sales;
```

-- Part F – Numeric and Date Functions --

-- Q76. Round all prices --

```
select round(price) as rounded_price from sales;
```

-- Q77. Apply CEIL() on prices --

```
select ceil(price) as ceil_price from sales;
```

-- Q78. Apply FLOOR() on prices. --

```
select floor(price) as floor_price from sales;
```

-- Q79. Find MOD(quantity,2) --

```
select quantity, MOD(quantity,2) as reminder from sales;
```

-- Q80. Find MOD(quantity,3) --

```
select quantity, MOD(quantity, 3) as reminder from sales;
```

```
-- Q81. Display the current date.  --
   select current_date();

-- Q82. Display the current time --
  select current_time;

-- Q83. Display the current timestamp.--
   select current_timestamp;

-- Q84. Extract the year from order_date --
select order_date,year(order_date) as order_date from sales;

-- Q85. Extract the month from order_date --
select order_date,month (order_date) as order_month from sales;

-- Q86. Extract the day from order_date. --
 select order_date, day(order_date) as order_day from sales;

-- Q87. Display orders placed in January --
 select * from sales where month(order_date) = 1;

-- Q88. Display orders placed before 20 January 2026.--
 select * from sales where order_date< '2026-01-20';
```

```
-- Q89. Display orders placed after 20 January 2026. --
  select * from sales where order_date>' 2026-01-20';

-- Q90. Display the latest order date --
  select min(order_date) as latest_order_date from sales;

-- Part G – UPDATE and DELETE --
 -- Q91. Update the city of Akash.--
  update sales set city = 'pune' where customer_name = 'akash';

-- Q92. Update the rating of Rahul.--
 update sales set rating = '4.2' where customer_name = 'rahul';

-- Q93. Change all Card payments to UPI.
  update sales set payment_mode = 'UPI' where payment_mode = 'card' ;

-- Q94. Increase all product prices by 500. --
  update sales set price = price + 500 ;

-- Q95. Decrease all product prices by 100 --
  update sales set price = price - 100 ;

-- Q96. Change category Electronics to Gadgets. --
  update sales set category = 'Gadgets' where category = 'Electronics';
```

```
-- Q97. Change city Pune to Nagpur. --
  UPDATE sales SET city = 'Nagpur' WHERE city = 'Pune';

-- Q98. Update the quantity of a selected product. --
  update sales set quantity = 1 where sale_id = 2 ;

-- Q99. Update a product name.--
   update sales set product_name = 'sandles' where sale_id = '20';

-- Q100. Update a customer name --
   update sales set customer_name = 'poonam' where sale_id = '19';

-- Q101. Delete a specific customer record. --
   delete from sales where sale_id = '1';

-- Q102. Delete all Sports category records --
   delete from sales where category = 'sports';

-- Q103. Delete records with rating below 4.0 --
   delete from sales where rating <= 4.0;

-- Q104. Delete all Cash payment records --
          delete from sales where payment_mode = 'cash';
-- Q105. Delete records from Jaipur. --
          delete from sales where city = 'jaipur';
         SET SQL_SAFE_UPDATES = 0;
```

-- Part H – GROUP BY --

-- Q106. Display city-wise order count --

```
select count(city) from sales;
```

```
SET SQL_SAFE_UPDATES = 1;
delete from sales where city = 'jaipur';
```

-- Q107. Display category-wise order count. --

```
select category,count(*) as order_count from sales group by category;
```

-- 108. Display payment mode-wise order count --

```
select payment_mode, count(*) as order_count from sales group by payment_mode;
```

-- Q109. Display gender-wise order count. --

```
select gender, count(*) as gender_count from sales group by gender;
```

--- Q110. Display city-wise total revenue. --

```
select city,sum(price*quantity) as totalrevenue from sales group by city;
```

--- Q111. Display category-wise total revenue--

```
select category, sum(price*quantity) as categotywiserevenue from sales group by
category;
```

--- Q112. Display payment mode-wise total revenue --

```
select payment_mode,sum(price*quantity) as modewiserevenue from sales group by
payment_mode;
```

--- Q113. Display city-wise average revenue. --

```
select city, avg(price*quantity) as citywiseavgrevenue from sales group by city;
```

--- Q114. Display category-wise average revenue. ---

```
select category, avg(price*quantity) as categorywiserevenue from sales group by category;
```

--- Q115. Display city-wise maximum sale ---

```
select city, max(price*quantity) as maxsale from sales group by city;
```

--- Q116. Display city-wise minimum sale. ---

```
select city, min(price* quantity) as minsale from sales group by city;
```

--- Q117. Display category-wise maximum sale. --

```
select category, max(price*quantity) as maxsale from sales group by category;
```

--- Q118. Display category-wise minimum sale ---

```
select category,min(price* quantity) as minsale from sales group by category;
```

--- Q119. Display city and category-wise revenue ---

```
select city,category, sum(price*quantity) as citycategorywiserevenue from sales group by city,category;
```

--- Q120. Display city and payment mode-wise revenue. --

```
select city,payment_mode, sum(price*quantity) as citypayment_mode from sales group by city,payment_mode;
```

-- Part I – HAVING Clause

--- Q121. Display cities having total revenue greater than 50,000. ---

```
select city,sum(price*quantity) as totalrevenue from sales group by city having
sum(price*quantity)>50000;
```

```
show databases;

use retail_store_db;

show tables;

desc sales;
```

-- Q122. Display categories having total revenue greater than 60,000. --

```
select category, sum(price*quantity) as  salesrevenue from sales group by category having
sum(price*quantity)>60000;
```

-- Q123. Display cities having average sale greater than 10,000. --

```
select city, avg(price*quantity) as avgsale from sales group by city having
avg(price*quantity)>10000;
```

--- Q124. Display categories having average rating greater than 4.5 --

```
select category, avg(rating) as avgrating from sales group by category having
avg(rating)>4.5;
```

-- Q125. Display cities having more than 3 orders --

```
select city, count(*) as more_order from sales group by city having count(*)>3;
```

-- Q126. Display categories having more than 2 orders --

```
select category, count(*) as most_order from sales group by category having count(*)>3;
```

-- Q127. Display payment modes having more than 5 orders --

```
select payment_mode, count(*) as most_payment_mode from sales group by
payment_mode having count(*)>3;
```

--- Q128. Display cities having maximum sale greater than 40,000 --

```
select city, sum(price*quantity) as max_sale from sales group by city having
sum(price*quantity)>40000;
```

--- Q129. Display categories having minimum sale greater than 1,500. --

```
select category, sum(price*quantity) as min_sale from sales group by category having
sum(price*quantity)<1500;
```

-- Q130. Display cities having average age greater than 25 --

```
select city, avg(age) as greaterage_age from sales group by city having avg(age) >25;
```

-- Part J – WHERE + GROUP BY + HAVING

-- Q131. Display city-wise revenue for UPI transactions.

```
select city , sum(price*quantity) as revenue from sales where payment_mode = 'upi' group
by city;
```

-- Q132. Display city-wise revenue for Electronics products. --

```
select city, sum(price*quantity) as revenue from sales where category = 'electronics'
group by city;
```

-- Q133. Display city-wise revenue generated by female customers. --

```
select city, sum(price*quantity) as revenue from sales where gender = 'female' group by
city;
```

-- Q134. Display city-wise revenue for orders above 5,000. --

select city, sum(price*quantity) as citywise_revenue from sales where (price*quantity)> 5000 group by city;

-- Q135. Display cities where revenue from orders above 5,000 exceeds 50,000. --

select city, sum(price*quantity) as cities_revenue from sales where(price*quantity)>5000 group by city having sum(price*quantity)>50000;

-- Q136. Display categories with more than 2 UPI orders --

select category,payment_mode as more_ordermode from sales where payment_mode ='upi' group by category having count(*)>=2;

--- Q137. Display categories where Card payment revenue exceeds 10,000.--

SELECT category, SUM(price * quantity) AS Revenue from sales WHERE payment_mode = 'Card'GROUP BY category HAVING SUM(price * quantity) > 10000;

show databases;

use retail_store_db;

--- Q138. Display category-wise count of products having rating above 4.5 --

select category, count(product_name) as rating from sales where rating > 4.5 group by category;

-- Q139. Display city-wise revenue generated by customers older than 25.--

select city,sum(price*quantity) as total_revenue from sales where age>25 group by city;

-- Q140. Display cities where revenue generated by customers older than 25 exceeds 20,000. ---

```
select city, sum(price*quantity) as older_revenue from sales where age > 25 group by city having sum(price*quantity)> 20000;
```

-- Final Business Reports --

-- Q141. Identify the city generating the highest revenue. --

```
select city,sum(price*quantity) as highest_revenue from sales group by city order by highest_revenue desc limit 1;
```

-- Q142. Identify the category generating the highest revenue --

```
select category,sum(price*quantity) as category_revenue from sales group by category order by category_revenue desc limit 1;
```

-- Q143. Find the most frequently used payment method --

```
select payment_mode, count(*)  as usage_count from sales group by payment_mode order by usage_count desc limit 1;
```

-- Q144. Find the best-rated category. --

```
select rating, count(*) as best_rated from sales group by rating order by best_rated desc limit 1;
```

-- Q145. Find the highest spending customer --

```
select customer_name, sum(price*quantity) as high_spent_cust from sales group by customer_name order by high_spent_cust desc limit 1;
```

-- Q146. Find the most popular product. --

select product_name, sum(quantity) as popular_product from sales group by product_name order by popular_product desc limit 1;

-- Q147. Display the top 3 cities by revenue. --

select city, sum(price*quantity) as cities_revenue from sales group by city order by cities_revenue desc limit 3;

-- Q148. Display the top 3 categories by revenue. --

select category, sum(price*quantity) as category_revenue from sales group by category order by category_revenue desc limit 3;

-- Q149. Generate a monthly sales summary report. --

select month(order_date) as month, sum(price*quantity) as total_revenue,count(distinct sale_id) as total_orders from sales group by(order_date) order by month;

-- Q150. Generate a complete business performance report using all concepts learned so far. --

```
select month(order_date) as month,
    sum(price*quantity) as total_revenue,
    count(order_id) as total_orders,
    count(distinct customer_id) as unique_customers,
    (select product_name from sales
    group by product_name,
    order by sum(quantity desc limit 1) as most_popular_product,
    (select payment_mode from sales group by payment_mode)
    );
```